

Chapter 14

Suggestion Scheme/Quality Circles

14.1 Suggestion Scheme

The Scheme provides an opportunity for promoting creativity through constructive thinking, recognition for individual ingenuity and chance to participate in the development of the company.

Definition of 'suggestion'

(i) All ideas concerning the following aspects pertaining to the Plant are acceptable as suggestions:

- a) Reduction of cost, waste, spillage, maintenance, hazards and possibilities of accidents
- b) Increase of utility, quality, yield or output of products.
- c) Conservation of materials, energy, power, time on process
- d) Improvement of the product or its design.
- e) Rationalisation of work, materials, or methods.
- f) Simplification of practices, procedures and processes.
- g) Improvement in advertising and sale of products or new sources of revenue.
- h) Improvement in civic problems traffic, hygiene and cleanliness.

(ii) The following aspects are however outside the scope of the scheme:

- a) Matters concerning organization.
- b) Matters within the sphere of Industrial Relations and collective bargaining.
- c) Replacement of facilities such as machines tools and other machinery and equipments.
- d) Extension of existing practices and equipments of the Plant.
- e) Items to which the management has already given consideration and on which action is pending, postponed or abandoned.
- f) Company's policy matters.
- g) Any other matter decided by the management.

Processing of Suggestion

The awards may be a combination of any of the following:

- (a) Cash award.
- (b) Token Gift.
- (c) Letter of appreciation from a senior officer/HOD.
- (d) Certificate of merit from Head of Works or CEO/Head of the Unit.

Wide publicity is given to the deserving suggestions through different agencies. The deserving cases are referred for coveted awards such as Prime Minister's Shram Awards & Viswakarma Rashtriya Puraskar etc.

The benefits for the organization are:

- i. Development of coherent problem solving methodology.
- ii. Increased production / productivity
- iii. Enhanced motivation
- iv. Improved quality
- v. Better employer - employee relationship

vi. Improved employee involvement.

14.2 BUSINESS EXCELLENCE (BE)

Business Excellence (BE) encompasses proven and outstanding practices for management of organisations and achievement of continuous improvement of processes, systems and organisational culture. BE initiatives can be applied to all kinds of businesses whether private or public, for-profit or not-for-profit and small or large.

Over the years there have been many proven practices for organisational improvement like Total Quality Management (TQM), Six Sigma, Business Improvement, Process Improvement, Business Transformation and Lean Manufacturing etc. All these come under the purview of the broad term “Business Excellence”.

In today’s dynamic, competitive and volatile world, organisations aiming for excellence can cater best to the needs of the stakeholders. This includes adopting systematic approach towards process optimization, meeting customer needs & regulatory requirements.

Applying Business Excellence in manufacturing companies can improve operational practices, reduce costs, minimise wastages, improve quality, yield and on-time deliveries. This naturally translates to growth in sales, revenues and profits. On the non-tangible side, BE facilitates better employee engagement and satisfaction, communication as well as adoption of innovative practices.

MANAGEMENT SYSTEM STANDARDS

A Management System is the way in which an organization manages the interrelated processes of the business in order to achieve its objectives. These objectives can relate to a number of different topics, including product or service quality, operational efficiency, environmental performance, health and safety in the workplace and many more.

ISO (International Organisation for Standardisation) is an independent, non-governmental organisation with a membership of 170 national standard bodies including BIS (Bureau of Indian Standards). ISO is responsible for framing International Standards. It was established in 1947 and has its headquarters in Geneva, Switzerland.

ISO Standards are internationally agreed by Experts. ISO Standards are not Product or Service standards, they are Process standards and can be used by product manufacturers and service providers. These standards cover a huge range of activities covering industries from technology to food safety to agriculture and healthcare. ISO has developed over 25000 International Standards and has 825 Technical committees and subcommittees to take care of standards development.

The Advantages of having ISO Standards:

- 1) Help businesses of any size and sector to reduce costs, increase productivity and access new markets

- 2) Development of national and international regulation
- 3) Reduce barriers to international trade
- 4) Creating a Continuous cycle of self-evaluation & correction

PDCA CYCLE

PDCA (Plan-Do-Check-Act) Cycle can be applied to all processes and management system standards as a whole.

It is a business methodology focusing on continuous improvement. Originally developed by Dr. W. Edwards Deming, the PDCA cycle is also known as the Deming cycle. It acts as a guiding principle for organizations to achieve ongoing growth and success.

The four steps in PDCA Cycle include:

1. **Plan** : Set objectives and build processes necessary to deliver results
2. **Do**: Implement what was planned
3. **Check**: Monitor and measure processes and results against the objectives
4. **Act**: Take actions to improve results

PDCA Approach helps organisations in systematically planning, executing, evaluating and refining processes to enhance efficiency and continuously improve the performance of an organization.

STRUCTURE OF ISO STANDARDS

High Level Structure (HLS) is a common structure established for all Management System Standards that are framed by the International Organization for Standardization (ISO).

The High-Level Structure or HLS is a set of 10 clauses as below:

- 1) Scope
- 2) Normative references
- 3) Terms and definitions
- 4) Context of the organization
- 5) Leadership
- 6) Planning
- 7) Support
- 8) Operation
- 9) Performance evaluation
- 10) Improvement

The advantage of HLS is that all ISO standards have a uniform set of clauses which helps in greater integration of multiple standards within one organization and they are easier to read and understand.

Integrated Steel Plants in SAIL are certified to the following ISO Standards (Either all or a selected few):

- ISO 9001:2015 Quality Management System(QMS)
- ISO 14001:2015 Environmental Management System(EMS)
- ISO 45001:2018 Occupational Health & Safety Management System(OHSMS)
- ISO 50001:2018 Energy Management System(EnMS)
- ISO 27001:2013 Information Security Management System(ISMS)
- ISO 37001:2016 Anti-Bribery Management System (ABMS)

Important Non-ISO Standards adopted by SAIL integrated plants include:

- SA 8000:2014 - Social Accountability Standard established by Social Accountability International (SAI). It is a framework for organizations of all types to conduct business in a way that is fair for workers and which demonstrates their adherence to the highest social standards.
- CE Mark Certification - Required for Exporting products to the European Market.

Implementing ISO Standards involves audits by external certification bodies. Recertification audits are held every three years and surveillance audits are organised annually.

14.3 QUALITY CIRCLES (QC)

Quality Circles are an effective way to improve quality and productivity through participation of the employees within the organisations. A Quality Circle is a small group of employees (4-6) who volunteer to help in solving departmental problems with limited resources. This small group utilizes the QC concept and techniques, display their creativity for self and mutual development. The various benefits from QCs include personality development, positive work environment, increased productivity, team spirit and cohesiveness.

The concept of quality circles first originated in Japan in the early 1960s, following a post-war reconstruction period during which the Japanese placed a great deal of emphasis on improving and perfecting their quality control techniques. As a direct result of work carried out to train foremen during that period, the first quality circles were conceived and the first three circles registered with the Japanese Union of Scientists and Engineers (JUSE) in 1962. In India, the concept of quality circles has taken nearly twenty years to reach India after its birth in Japan in 1962. The Bharat Heavy Electrical Limited (BHEL) was the first to launch quality circles in 1980.

The Twelve Steps of Problem Solving in QC are:

- Step 1- Identification of Problem
- Step 2- Selection of Problem
- Step 3- Define the Problem
- Step 4- Analysis of the Problem
- Step 5- Identification of Causes
- Step 6- Finding out root Causes
- Step 7- Data Analysis

Step 8- Developing Solution
Step 9- Foreseeing Probable Resistance
Step 10- Trial Implementation and Check Performance
Step 11- Regular Implementation
Step 12- Follow up Review

Quality Circles use various problem solving techniques viz: Flow Diagram, Brainstorming, Data Collection, Graphs, Stratification, Pareto Diagram, Cause & Effect Diagram, Scatter Diagram, Histogram and Control Charts in above stated 12 Steps for solving work related problems.

All integrated steel plants in SAIL have active QCs that have brought laurels for the company at National level(NCQC – National Convention on Quality Concepts) & International level(ICQCC - International Convention on Quality Control Circles) competitions
